Optimal Field Sampling Framework for Disease Surveillance of Clonally Propagated Plants

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ABSTRACT

Vegetatively propagated crops are vulnerable to systemic pathogen accumulation, yet surveillance strategies often fail to capture the interplay between latent infection dynamics, spatial dispersal, and propagation practices. We develop a spatially explicit delay-differential compartmental model that integrates within-farm transmission, transboundary dispersal, and cutting-based propagation. Analytical results identify two critical tipping points in infectious burden: a local persistence threshold ($\theta_1 = 0.1$) and a spatial export threshold ($\theta_2 = 22.48$), revealing a transition zone for local containment before a farm becomes a regional risk exporter. We find that infections surge after the first replanting cycle, driven by a "Latency Trap" where long incubation delays (τ) mask a large latent infection reservoir. Embedding the model into a Bayesian decision-theoretic framework reveals that optimal surveillance effort increases with epidemic establishment, ranging from a baseline detection effort at onset ($n = 50$) to expanded surveillance under larger budgets ($n = 138$). Budget sensitivity analysis further shows that surveillance effort progressively shifts toward broader spatial coverage as epidemics mature. The spatial information surface demonstrates that the informational value of sampling is concentrated near epidemiological transition zones rather than being uniformly distributed across the landscape. These findings highlight that early detection in clonal crops requires front-loaded sampling effort, followed by intensified diagnostic precision as the pathogen reaches endemicity. This coupled dynamical-statistical framework provides both theoretical insights into the invisibility of latent threats and operational tools for optimizing resource allocation in national plant health programs.

HIGHLIGHTS

- Mechanistic model identifies a transition between local persistence ($\theta_1 = 0.1$) and regional export potential ($\theta_2 = 22.48$).
- The "Latency Trap" generated by long incubation delays masks large latent infection reservoirs prior to widespread symptom expression.
- Budget sensitivity analysis indicates that optimal surveillance effort increases with epidemic establishment and resource availability.
- Accounting for within-farm sampling dependence shifts optimal surveillance from intensive local sampling toward broader regional coverage at higher budgets.
- Spatial information weighting suggests that sampling near epidemiological transition zones is more informative than uniform random allocation.

1. INTRODUCTION

Pests and diseases are among the most pressing threats to ongoing global efforts to achieve sustainable agriculture and address zero hunger worldwide. Specifically, losses incurred as a direct result of the impact of pests and diseases on crops, including tubers, roots, and bananas, have been estimated to be, on average,

17.2% for potatoes, 24% for cassava, and 29% for bananas (Szyniszewska and et al. 2017; Okonya and et al. 2019; Savary et al. 2019; Carvajal-Yepes et al. 2019). Tubers, roots, and bananas are integral to food security and are propagated vegetatively. They are particularly susceptible to disease because pathogens can persist and spread throughout the growing seasons. This is because farmers often source seeds from seemingly healthy plants. In many developing countries, seed exchange remains the primary means of acquiring planting material from one season to the next (Almekinders et al. 2019).

Effectively tackling these threats requires surveillance systems; yet, current approaches in plant health face challenges of ad hoc sampling strategies, limited resources, and difficulty balancing statistical power with costs and logistics. These challenges hinder adequate sampling at different spatial and temporal scales at the onset of, and in monitoring, the progression of the epidemic. Recent studies offer insights into optimizing surveillance strategies to address these issues. When resources are limited, a surveillance program under budget constraints is suggested, determining the intensity and frequency of inspections and sampling (Lázaro et al. 2021). The authors developed a sequential adaptive delimiting survey with increasing spatial resolution for *Xylella fastidiosa* in Alicante, Spain. Optimization was carried out through simulation, and sampling intensity thresholds were evaluated based on their effects on incidence estimation. Although this method offers a structure for estimating samples, its precision is insufficient to reflect the vertical and horizontal transmission dynamics, as in mechanistic models. This case study demonstrates the majority of several sample size estimation methods, which are often not rooted in epidemiological theory, resulting in difficulties with adaptability and making decisions based on information. The integration of epidemiological perspectives has been shown to improve sampling initiatives, thereby facilitating a more efficient identification of emerging pathogens (Parnell et al. 2017).

Mathematical models elucidate disease mechanisms in vegetatively propagated crops and evaluate the effectiveness of disease management strategies. Deterministic compartmental models, including susceptible, infected, and removed, are employed to analyze the transmission dynamics of cassava mosaic disease, which spreads through vectors and contaminated planting materials in cassava farms (Chapwanya and Dumont 2021). Specifically, the authors determined conditions for local (global) asymptotic stability and demonstrated that Hopf Bifurcation may occur when roguing is used to control the epidemics.

The unpredictable patterns of crop outbreaks associated with regional environmental factors was captured through isotropic kernel dispersal and stochastic processes, in which the transmission probabilities decay along all radial directions as one moves farther from the epidemic front (Fabre et al. 2021). This framework was implemented to assess different cassava brown streak disease (CBSD) management options and optimize disease surveillance protocols by examining the effects of survey timing, sampling intensity, and diagnostic accuracy on CBSD detection and incidence estimation (Ferris et al. 2020). They found that a minimum of 30 plants need to be surveyed within a farm, approximately 299 days post-planting, to obtain a minimum accuracy of 50% precision in identifying CBSD. However, the accuracy of the CBSD incidence within a farm remains challenging since its estimation depends primarily on survey accuracy rather than the number of plants surveyed or the timing of the survey. Moreover, for the successful early identification of CBSD, assessments must be conducted in areas lacking previous disease documentation, and the number of farms per region (per-region sample size) should align with the density of cassava farms (Ferris et al. 2024). Even though these previous studies explored and proposed a sampling estimation on the onset of an epidemic, their application

is still constrained by the thorough definition of the host landscape. They fall short of establishing a universal framework applicable to other vegetatively propagated crops at the onset and monitoring of epidemics.

Moreover, many crop disease surveillance programs, especially in developing countries, operate under budget constraints, limiting the intensity and frequency of sampling. This necessitates sampling optimization strategies to maximize efficiency within these limitations while balancing the trade-off between statistical power, costs, and logistical constraints (Barrett et al. 2010; Lázaro et al. 2021). Although optimizing sampling offers promising solutions, their implementation must consider local contexts and pathogen-specific dynamics.

We introduce a decision-theoretic framework for sampling optimization that accounts for the hidden burden of latent infection. Specifically, we develop a spatially explicit delay differential model of disease spread in clonally propagated plants, capturing the critical decoupling between symptomatic observation and true infectious burden. We then derive a transmission-based utility function that links these mechanistic dynamics to the Expected Value of Sample Information (EVSI), acknowledging that the marginal value of data peaks near critical persistence and export thresholds. Finally, we establish an optimization framework to determine sample sizes that adapt to the epidemic phase.

2. METHODOLOGY: DYNAMICAL MODEL AND SAMPLING FRAMEWORK

This section presents a spatially explicit delay differential compartmental model describing the transmission dynamics of a clonal plant pathogen across multiple farms. Local transmission, spatial dispersal, and human-mediated propagation through cuttings are integrated within a population-conserving framework.

2.1. Model Structure and Assumptions

We model disease spread in vegetatively propagated cropping systems as the result of: (i) local transmission within farms, (ii) spatial transmission between farms, and (iii) human-mediated transmission through the exchange of planting material.

Transmission via cuttings is treated as a recruitment mechanism introducing infected plants into the system, rather than vertical transmission in the epidemiological sense.

We consider a meta-population system of M farms indexed by $j = 1, \ldots, M$. Each farm contains a closed and constant plant population

$$N_j = S_j(t) + L_j(t) + I_j(t),$$

with population conservation enforced through symmetric removal and recruitment.

Plants are classified into three epidemiological compartments:

- $S_j(t)$: susceptible plants;
- $L_j(t)$: latent infected plants (infected, asymptomatic, but infectious);
- $I_j(t)$: infectious plants (symptomatic).

In plant pathology, latency corresponds to an asymptomatic infectious stage. Accordingly, latent plants contribute to transmission. Progression from latent to symptomatic infection occurs after a fixed delay $\tau > 0$, representing the incubation period.

2.1.1. Force of Infection: Local and Spatial Components

The total force of infection acting on farm j at time t is

$$\Lambda_j(t) = \beta \frac{L_j(t) + I_j(t)}{N_j} + \sum_{k \neq j} \alpha \frac{L_k(t) + I_k(t)}{N_k} e^{-\delta d_{jk}},$$

where β and α denote local and between-farm transmission rates, δ controls distance decay, and d_{jk} is the distance between farms j and k.

This formulation preserves mass-action transmission while remaining density-dependent.

2.1.2. Human-Mediated Propagation via Cuttings

A fraction γ of plants is replaced per unit time through replanting with locally sourced cuttings. The prevalence of symptomatic infection is

$$\psi_j(t) = \frac{I_j(t)}{N_j}.$$

The observable symptomatic prevalence differs from the true infectious burden:

$$\theta_j(t) = \frac{L_j(t) + I_j(t)}{N_j}.$$

While $\psi_j(t)$ governs observability during field surveys, $\theta_j(t)$ governs epidemic transmission. The discrepancy between these quantities is induced by the incubation delay τ and defines the hidden latent reservoir.

Cuttings are selected with a behavioral response modeled by the saturation function

$$f(\psi_j) = \frac{\psi_j}{1 + \kappa \psi_j},$$

where $\kappa \geq 0$ quantifies disease awareness.

Recruitment flows are:

- susceptible recruitment at rate $\gamma N_j (1 - f(\psi_j(t)))$,
- latent recruitment at rate $\gamma N_j f(\psi_j(t))$,
- symmetric removal at rate γ from all compartments.

2.1.3. Complete Delay Differential Model

Let

$$\Phi_j(t) = \Lambda_j(t) S_j(t) + \gamma N_j f(\psi_j(t))$$

denote the total rate at which plants enter the latent compartment via infection or infected cuttings.

The full system for farm j is

$$\frac{dS_j}{dt} = -\Lambda_j(t) S_j(t) + \gamma N_j \big(1 - f(\psi_j(t))\big) - \gamma S_j(t), \tag{1}$$

$$\frac{dL_j}{dt} = \Phi_j(t) - \Phi_j(t - \tau) e^{-\gamma \tau} - \gamma L_j(t), \tag{2}$$

$$\frac{dI_j}{dt} = \Phi_j(t - \tau) e^{-\gamma \tau} - \gamma I_j(t). \tag{3}$$

Model properties:

- All plants entering latency (via infection or cuttings) progress to symptomatic infection after delay τ, provided they are not removed.
- The survival factor $e^{-\gamma \tau}$ accounts for removal during latency.

- No infected plants are trapped in the latent class.
- Total population is conserved:

$$\frac{dN_j}{dt} = 0 \quad \forall j.$$

The deterministic delay system defined in Section 2 provides the expected epidemic trajectory conditional on fixed parameters and initial conditions. However, surveillance decisions are inherently made under uncertainty arising from stochastic transmission, imperfect knowledge of parameters, and incomplete observations. To formally connect the mechanistic epidemic model with Bayesian decision theory, we introduce a probabilistic epidemic layer.

2.2. Latent Epidemic State

Let

$$\Theta(t) = \{\theta_j(t)\}_{j=1}^{M}, \qquad \theta_j(t) = \frac{L_j(t) + I_j(t)}{N_j},$$

denote the vector of true infectious burdens across the metapopulation.

Rather than treating $\Theta(t)$ as deterministic, we model it as a latent random field:

$$\Theta(t) \sim p(\Theta(t) \mid \eta),$$

where

$$\eta = (\beta, \alpha, \delta, \tau, \gamma, \kappa)$$

collects uncertain epidemiological parameters.

The deterministic DDE system therefore defines the conditional mean trajectory

$$\mathbb{E}[\Theta(t) \mid \eta].$$

2.3. Parameter Uncertainty

To propagate uncertainty into surveillance decisions, parameters are assigned prior distributions:

$$\beta \sim p(\beta),$$
$$\alpha \sim p(\alpha),$$
$$\delta \sim p(\delta),$$
$$\tau \sim p(\tau),$$
$$\gamma \sim p(\gamma).$$

The prior distribution of the epidemic state is then induced through the DDE dynamics:

$$p(\Theta(t)) = \int p(\Theta(t) \mid \eta)\, p(\eta)\, d\eta.$$

2.4. Biological and Observable Prevalence

A critical distinction must be made between biological infection and observable infection.

2.4.0.1. Biological prevalence. A farm is biologically infected whenever latent or symptomatic infection is present:

$$\pi_B(t) = \frac{1}{M} \sum_{j=1}^{M} \mathbf{1}\big(L_j(t) + I_j(t) > 0\big)\,.$$

2.4.0.2. Observable prevalence. A farm is observable only through symptomatic infection:

$$\pi_O(t) = \frac{1}{M} \sum_{j=1}^{M} \mathbf{1}\big(I_j(t) > \epsilon\big) .$$

The discrepancy

$$\pi_B(t) - \pi_O(t)$$

quantifies the hidden latent reservoir generated by the incubation delay τ and formally defines the "Latency Trap."

Interpretation. The deterministic DDE trajectory therefore represents the conditional epidemic expectation, while uncertainty around the trajectory defines the Bayesian prior required for EVSI calculations.

2.4.1. Initial History Functions

Because the system is a delay differential equation, solutions require initial history functions

$$(S_j(\theta), L_j(\theta), I_j(\theta)), \quad \theta \in [-\tau, 0],$$

which represent the epidemiological state prior to the start of observation.

These histories encode the pre-existing infections necessary for delayed progression and are required for mathematical well-posedness but do not introduce additional model assumptions.

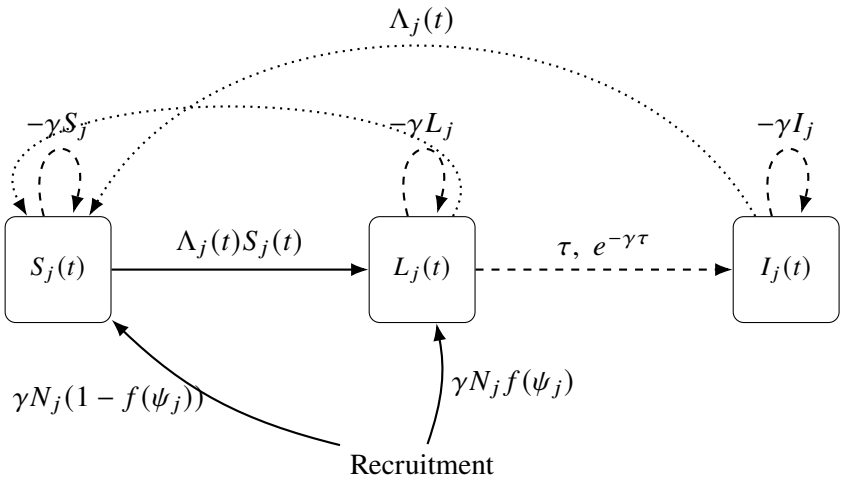


Figure 1. Schematic representation of the delay differential epidemic model for farm j. Susceptible plants S_j become latently infected at rate $\Lambda_j(t)S_j(t)$, where both latent and infectious plants contribute to transmission. All latent plants (including those originating from infected cuttings) progress to symptomatic infection after a fixed delay τ, provided they are not removed, with survival probability $e^{-\gamma\tau}$. Population size is conserved through symmetric replacement at rate γ.

2.4.2. The Latency Trap and Sampling Rigor

A feature of this model is the disconnect between biological risk and observability. As shown in Fig. 1, only individuals in the $I_j(t)$ compartment are symptomatic and thus detectable during field surveys. However, the transmission pressure $\Lambda_j(t)$ is driven by both $I_j(t)$ and the invisible latent class $L_j(t)$.

This "Latency Trap" is the primary driver for the high sampling effort (m, n) identified in our results. Because $L_j(t)$ can represent a hidden reservoir, especially during the onset and endemic phases; the Bayesian update requires a sample size m, n (number of farms and plants per farm, respectively) to ensure that the posterior estimate of regional risk $\pi(t)$ accounts for the progression delay τ. Without this link, the sampling framework would risk underestimating the epidemic trajectory.

3. MATHEMATICAL ANALYSIS OF THE MODEL

We analyze the well-posedness, positivity, invariant region, disease-free equilibrium (DFE), and the basic reproduction number $\mathcal{R}_0$ of the delay differential system. For clarity, the farm index j is omitted when no ambiguity arises.

3.1. Well-posedness of the Delay System

Let $\tau > 0$ be fixed. The state variables

$$(S(t), L(t), I(t)) \in \mathbb{R}^3_+$$

satisfy a system of delay differential equations with discrete delay τ. Define the history space

$$C := C([-\tau, 0], \mathbb{R}^3),$$

and assume a nonnegative initial history

$$(S(\theta), L(\theta), I(\theta)) = \varphi(\theta) \geq 0, \quad \theta \in [-\tau, 0].$$

The right-hand side of the system is locally Lipschitz continuous in (S, L, I) and continuous with respect to the delayed arguments. Moreover, the force of infection $\Lambda(t)$ and the recruitment function $f(\psi(t))$ are smooth for $S, L, I \geq 0$.

Proposition 1 (Existence and Uniqueness).
For any nonnegative initial history $\varphi \in C$, there exists a unique solution

$$(S(t), L(t), I(t)) \in C([-\tau, T), \mathbb{R}^3_+)$$

for some $T > 0$.

Proof.
This follows from standard theory for functional differential equations with discrete delay (Hale and Verduyn Lunel 1993), since the system is locally Lipschitz. □

3.2. Proposition: Positivity and Invariant Region

Positivity

Suppose the initial history satisfies

$$S(\theta), L(\theta), I(\theta) \geq 0, \qquad \theta \in [-\tau, 0].$$

Then the solution of the delay system remains nonnegative for all $t \geq 0$.

Proof. Using the variation-of-constants representation,

$$L(t) = L(0)e^{-\gamma t} + \int_{t-\tau}^{t} \Phi(s) e^{-\gamma(t-s)}\, ds,$$

where

$$\Phi(s) \geq 0.$$

Since the exponential kernel is nonnegative,

$$L(t) \geq 0.$$

Similarly,

$$I(t) = I(0)e^{-\gamma t} + \int_{0}^{t} \Phi(s-\tau) e^{-\gamma\tau} e^{-\gamma(t-s)}\, ds,$$

which implies

$$I(t) \geq 0.$$

Finally, if $S(t) = 0$, then

$$\frac{dS}{dt} = \gamma N(1 - f(\psi(t))) \geq 0.$$

Therefore trajectories cannot leave the nonnegative orthant. □

3.3. Population Conservation and Invariant Region

Summing the three equations yields

$$\frac{d}{dt}(S + L + I) = \gamma N - \gamma(S + L + I).$$

Since N is constant and the initial condition satisfies

$$S(0) + L(0) + I(0) = N,$$

it follows that

$$S(t) + L(t) + I(t) = N \quad \forall t \geq 0.$$

Proposition 3 (Invariant Region).

The set

$$\Omega = \left\{(S, L, I) \in \mathbb{R}^3_+ : S + L + I = N\right\}$$

is positively invariant and attracts all solutions.

□

3.4. Disease-Free Equilibrium (DFE)

At the disease-free equilibrium,

$$L^* = I^* = 0,$$

implying $\psi^* = 0$ and $f(0) = 0$. The susceptible equation reduces to

$$\frac{dS}{dt} = \gamma N - \gamma S,$$

with equilibrium

$$S^* = N.$$

Disease-Free Equilibrium:

$$E_0 = (S^*, L^*, I^*) = (N, 0, 0).$$

4. SPATIAL REPRODUCTION NUMBER

The basic reproduction number of the metapopulation system depends jointly on local transmission, delayed progression, and spatial coupling between farms.

4.1. Transmission Kernel

Define the spatial transmission matrix

$$K = (K_{jk})_{1 \le j,k \le M},$$

with entries

$$K_{jk} = \begin{cases} \beta, & j = k, \\ \alpha e^{-\delta d_{jk}}, & j \neq k. \end{cases}$$

The matrix K quantifies the expected force of infection generated from farm k toward farm j.

4.2. Expected Infectious Lifetime

An infected individual contributes to transmission during:

1. the latent infectious phase,
2. and the symptomatic infectious phase if surviving the delay.

The expected transmission-weighted infectious duration is

$$T_{\text{inf}} = \frac{1}{\gamma} + e^{-\gamma\tau}\frac{1}{\gamma}.$$

Thus,

$$T_{\text{inf}} = \frac{1 + e^{-\gamma\tau}}{\gamma}.$$

4.3. Metapopulation Reproduction Number

The next-generation operator is

$$\mathcal{K} = T_{\text{inf}}\, K.$$

The basic reproduction number is defined as the spectral radius:

$$\boxed{\mathcal{R}_0 = \rho(\mathcal{K}) = \frac{1 + e^{-\gamma\tau}}{\gamma}\,\rho(K).}$$

Interpretation. This formulation explicitly incorporates:

- delayed progression through τ,
- local transmission through β,
- spatial coupling through α,
- and metapopulation geometry through the spectral radius $\rho(K)$.

5. HIERARCHICAL OBSERVATION MODEL

The surveillance process is naturally hierarchical: plants are sampled within farms, and farms are sampled within the landscape. To preserve this structure, observations are modeled at the farm level rather than through a regionally aggregated Binomial approximation.

5.1. Plant-Level Detection

Suppose that n_j plants are sampled from farm j. Let

$$\psi_j(t) = \frac{I_j(t)}{N_j}$$

denote the symptomatic prevalence.

If σ is the diagnostic sensitivity for a symptomatic plant, then the probability of detecting at least one infected plant on farm j is

$$P_{d,j}(t) = 1 - (1 - \sigma\psi_j(t))^{n_j}.$$

When $n_j \ll N_j$, this expression approximates independent sampling with replacement.

5.2. Farm-Level Observation Model

Let

$$Y_j = \begin{cases} 1, & \text{if farm } j \text{ tests positive,} \\ 0, & \text{otherwise.} \end{cases}$$

Conditional on the latent epidemic state,

$$Y_j \mid \Theta(t) \sim \text{Bernoulli}(P_{d,j}(t)).$$

For a sampled set of farms $\mathcal{S}$, the likelihood becomes

$$\mathcal{L}(\Theta \mid Y) = \prod_{j \in \mathcal{S}} P_{d,j}(t)^{Y_j} \left(1 - P_{d,j}(t)\right)^{1-Y_j}.$$

5.3. Posterior Distribution

The posterior epidemic state is therefore

$$p(\Theta(t) \mid Y) \propto \mathcal{L}(\Theta \mid Y)\, p(\Theta(t)).$$

This formulation preserves:

- farm-level heterogeneity,
- spatial dependence,
- latent uncertainty,
- and the biological distinction between hidden and observable infection.

6. SAMPLING FRAMEWORK AND SURVEILLANCE UTILITY

6.1. Action Space, Decision Variables, and Transmission-Based Thresholds

Surveillance is decision-relevant only insofar as it alters management actions that modify epidemic trajectories. We therefore define a finite action space

$$\mathcal{A} = \{A_0, A_1, A_2\},$$

where each action corresponds to a qualitatively distinct intervention regime:

- A_0 (No intervention): routine monitoring only; unrestricted movement of planting material.
- A_1 (Local containment): farm-level quarantine, removal of symptomatic plants, and restriction of cuttings within farm j.
- A_2 (Regional firewall): coordinated intervention across farms, including movement bans and targeted culling, designed to interrupt spatial transmission.

Latent-inclusive infectious burden. Let

$$\theta_j(t) := \frac{L_j(t) + I_j(t)}{N_j}$$

denote the true infectious burden on farm j, which governs transmission. This quantity enters the force of

infection

$$\Lambda_j(t) = \beta\theta_j(t) + \sum_{k\neq j} \alpha\theta_k(t)e^{-\delta d_{jk}}.$$

Transmission-based management thresholds. We define two critical thresholds

$$0 < \theta_1 < \theta_2 < 1$$

that partition the action space.

Local persistence threshold (θ_1). The threshold θ_1 is defined as the minimum infectious burden such that local transmission exceeds replacement:

$$\beta\theta_1 = \gamma.$$

For $\theta_j < \theta_1$, infections are expected to die out locally; for $\theta_j \geq \theta_1$, farm j sustains endemic transmission.

Spatial export threshold (θ_2). The threshold θ_2 marks the transition from local risk to regional threat. Formally, θ_2 is defined as the smallest value of θ_j such that farm j becomes a net exporter of infection capable of triggering persistence in at least one neighboring farm:

$$\exists\, k \neq j \text{ such that } \alpha\theta_2 e^{-\delta d_{jk}} \geq \gamma - \beta\theta_k.$$

In words, θ_2 is reached when the spatial force of infection emitted by farm j is sufficient to push a neighboring farm k from subcritical ($\theta_k < \theta_1$) to supercritical transmission.

Action rule. Management decisions are therefore defined as

$$A(\theta_j) = \begin{cases} A_0, & \theta_j < \theta_1, \\ A_1, & \theta_1 \leq \theta_j < \theta_2, \\ A_2, & \theta_j \geq \theta_2. \end{cases}$$

The Latency Gap and Observability. During the epidemic growth phase, the symptomatic proportion $\phi(t) = I_j(t)/(L_j(t) + I_j(t))$ is less than unity. In the likelihood function, as the delay τ increases, the symptomatic prevalence $\psi_j \to 0$, causing the detection probability $P_d \to 0$ even if the true infectious burden $\theta_j \geq \theta_1$. This mathematical gap necessitates a compensatory increase in sample size n to resolve the "hidden" reservoir, a cost-driver we quantify in the following section.

The relationship between the observable prevalence ψ_j and the decision-relevant burden θ_j is governed by the time-varying symptomatic fraction $\phi(t)$:

$$\psi_j(t) = \phi(t) \cdot \theta_j(t), \quad \text{where } \phi(t) = \frac{I_j(t)}{L_j(t) + I_j(t)}. \tag{4}$$

Decision relevance of sampling. Let $\widehat{\theta}_j^{(n)}$ denote the posterior estimator of θ_j based on a sample of size n. Sampling has positive value only if it can alter the selected action:

$$\mathbb{P}\Big(A\big(\widehat{\theta}_j^{(n)}\big) \neq A(\theta_j)\Big) > 0.$$

Away from θ_1 and θ_2, additional samples do not change the decision and therefore have zero expected value.

Interpretation. Near θ_1, sampling protects against false negatives that allow silent, latent-driven persis-

tence. Near θ_2, sampling protects against catastrophic false negatives that permit spatial spillover and metapopulation-wide invasion. Thus, the sample size problem is inherently a problem of preventing incorrect action selection under spatial transmission risk, not of estimating prevalence with arbitrary precision.

Decision-Theoretic Linkage. The selection of an action $a \in \mathcal{A}$ is mediated by the observation X. While the DDE provides the prior state, the decision is updated using the regional Likelihood $\mathcal{L}(\pi|X, m, n)$ derived in the previous section (Eq. 8). The farm-level detection probability $P_d(\psi_j)$ bridges the plant-level effort (n) with the farm-level survey (m), ensuring that the management thresholds θ_1 and θ_2 are evaluated against a debiased posterior.

6.2. Bayesian Loss Function Driven by Metapopulation Risk

Surveillance decisions are evaluated through a Bayesian loss function that quantifies both the immediate economic cost of intervention and the future epidemiological damage resulting from delayed or insufficient control. Sampling is rational only insofar as it reduces expected loss near critical transmission thresholds.

Epidemic state. The relevant latent state on farm j is

$$\theta_j = \frac{L_j + I_j}{N_j},$$

which governs persistence, amplification, and spatial export of infection. Because θ_j is only partially observable, all decisions are based on the posterior distribution $\mathbb{P}(\theta_j \mid \text{data})$.

Total loss function. For an action $a \in \mathcal{A} = \{A_0, A_1, A_2\}$ and true state θ_j, we define

$$L(a, \theta_j) = L_{\text{epi}}(a, \theta_j) + L_{\text{econ}}(a, \theta_j),$$

where L_{epi} captures downstream epidemic damage and L_{econ} captures direct intervention costs.

Epidemiological Loss: Under-Intervention

Epidemiological loss arises when an action fails to suppress transmission in a regime where the metapopulation reproduction potential exceeds unity.

Define the effective reproduction pressure generated by farm j as

$$\mathcal{R}_j(\theta_j) = \frac{\beta}{\gamma}\left(1 + e^{-\gamma\tau}\right)\theta_j + \sum_{k \neq j} \frac{\alpha}{\gamma}\,\theta_j e^{-\delta d_{jk}},$$

which explicitly includes both local amplification and spatial export.

False negative loss. The epidemiological penalty is activated when the true state θ_j crosses the persistence threshold θ_1 but the observation x leads to a sub-optimal action (A_0 or A_1). For $\theta_j \geq \theta_1$:

$$L_{\text{FN}}(a, \theta_j) = C_{\text{epi}} \exp\left(\eta\left(\mathcal{R}_j(\theta_j) - 1\right)_+\right),$$

where $\mathcal{R}_j(\theta_1) = 1$ by definition of the local persistence threshold.

6.2.1. *Economic Loss: Over-Intervention*

Intervention costs depend on the spatial scale and intensity of the chosen action.

Local interventions (A_1). Destroying plants on farm j incurs

$$L_{\text{econ}}(A_1, \theta_j) = C_{\text{local}} \cdot N_j,$$

where C_{local} is the per-plant removal cost.

Regional interventions (A_2)**.** Regional culling acts as a "firewall" and targets farms in the neighborhood

$$\mathcal{N}(j) = \{k \mid d_{jk} < d_{\text{crit}}\},$$

where d_{crit} satisfies

$$e^{-\delta d_{\text{crit}}} = \varepsilon,$$

with ε a small cutoff threshold (e.g., 1%).

The economic loss is

$$L_{\text{econ}}(A_2, \theta_j) = C_{\text{regional}} \sum_{k \in \mathcal{N}(j)} N_k,$$

reflecting the wider impact of coordinated culling.

6.2.2. Neighbor State Dependence and Prior Expectation

A critical subtlety is that the decision threshold θ_2 for triggering A_2 depends on the infection states of neighbors θ_k via

$$\sum_{k \in \mathcal{N}(j)} \alpha \theta_j e^{-\delta d_{jk}} > \text{threshold},$$

implying that Farm j's spatial risk and optimal sampling size depend on uncertain θ_k values.

To resolve this, the metapopulation model provides *expected values* $\mathbb{E}[\theta_k]$ derived from current simulations or posterior distributions, enabling informed sampling designs that incorporate network-wide uncertainty.

6.2.3. Bayesian Loss Function in short

The total loss function captures a fundamental regime shift in surveillance strategy:

- When θ_j is far from thresholds θ_1 and θ_2, sampling effort m can be near zero because information has low marginal value.
- Near thresholds, sampling reduces uncertainty, balancing the high cost of false negatives with the economic cost of interventions.
- The metapopulation model drives this dynamic by quantifying spatial coupling and latent burden, enabling a context-dependen.

7. EXPECTED VALUE OF SAMPLE INFORMATION (EVSI)

The Expected Value of Sample Information (EVSI) at the farm level quantifies the reduction in decision loss achieved by sampling n plants. This is the critical scale where the "Latency Trap" is resolved.

7.1. Farm-Level Optimization (n)

For a specific farm j, the survey data x_j represents the number of symptomatic plants detected out of n sampled plants. The EVSI for farm j is defined as:

$$\text{EVSI}_j(m) = \min_{a \in \mathcal{A}} \mathbb{E}_{\theta_j}\left[L(a, \theta_j)\right] - \mathbb{E}_{x_j}\left[\min_{a \in \mathcal{A}} \mathbb{E}_{\theta_j | x_j}\left[L(a, \theta_j)\right]\right] \tag{5}$$

where the expectation $\mathbb{E}_{x_j}$ is taken over the distribution of survey outcomes $x_j \sim \text{Binomial}(n, \sigma\phi(t)\theta_j)$.

7.1.0.1. The Role of the Symptomatic Fraction. As the biological delay τ increases, the symptomatic fraction $\phi(t)$ decreases. To maintain a non-zero EVSI in high-latency regimes, the sample size n must increase to compensate for the "hidden" infectious burden θ_j.

7.2. Network-Level Allocation (m and n_j)

The regional coordinator optimizes the allocation of effort across the metapopulation. The decision variable is now the set of farms to visit (m) and the intensity of sampling at each (n_j).

7.2.0.1. Global Objective Function. The goal is to maximize the total Expected Net Gain of Sampling (ENGS) across all farms $\mathcal{J}$ under a global budget B:

$$\max_{\{n_j\}} \sum_{j \in \mathcal{J}} \left(\text{EVSI}_j(n_j) - C(n_j) \right) \quad \text{s.t.} \quad \sum_{j \in \mathcal{J}} \mathbb{I}(n_j > 0) \cdot (c_{\text{farm}} + n_j c_{\text{plant}}) \leq B \tag{6}$$

where $\mathbb{I}(n_j > 0)$ is an indicator function representing the fixed cost of visiting farm j, effectively defining the number of sampled farms $m = \sum \mathbb{I}(n_j > 0)$.

7.2.0.2. Regime Shift and Spatial Coupling. The metapopulation model informs the prior $P(\theta_j)$ used in the EVSI calculation.

- **High-Connectivity Spark Farms:** In Zone C (Refugium), if a farm has high connectivity ξ_j, the potential loss $L(a, \theta_j)$ is massive. This drives n_j^* higher to ensure the "Information Frontier" is monitored.
- **Saturated Core:** In Zone A, where θ_j is well above θ_2, the $\text{EVSI}_j(n)$ collapses to zero for all n, resulting in $n_j^* = 0$.

7.3. Sensitivity and Robustness Analysis

Finally, robustness is assessed with respect to key biological and economic parameters:

$$(\beta, \alpha, \delta, \tau, \gamma, c_0, c_1).$$

We examine the stability of n_j^* under parameter perturbations to identify the thresholds at which surveillance collapses or explodes, the parameters that most strongly control information value, and regimes where sampling recommendations are policy-robust.

8. SPATIAL OPTIMIZATION OF SAMPLE SIZE ALLOCATION

The objective of spatially optimized surveillance is not to answer the question *Where is the disease?"* but rather *Where will sampling most effectively alter regional management decisions?"*. Sampling effort is therefore allocated to maximize the expected reduction in metapopulation-wide loss under operational and budgetary constraints. Let $\theta_j(t) = \frac{L_j(t)+I_j(t)}{N_j}$ denote the infectious burden at farm j, and let $\Lambda_j^{\text{ext}}(t) = \sum_{k \neq j} \alpha\, \theta_k(t)\, e^{-\delta d_{jk}}$ be the external force of infection acting on farm j.

8.1. Hierarchical Allocation Logic: Solving for n^* and m^*

To resolve the "hidden burden" at the landscape scale, the optimization must solve for both n_j (plants per farm) and m (total farms). These are coupled via the Hierarchical Observation Model:

$$P_d(\theta_j, \tau) = 1 - (1 - \sigma \cdot \phi(t) \cdot \theta_j)^n \tag{7}$$

where σ is diagnostic sensitivity and $\phi(t)$ is the symptomatic fraction derived from the DDE.

Optimal Farm-Level Effort (n_j^*). At the farm level, n_j is optimized to maximize the farm-specific ENGS:

$$n_j^* = \arg\max_{n \in \mathbb{N}_0} \left(\text{EVSI}_j(n) - (c_{\text{fixed}} + c_{\text{plant}} n) \right)$$

In early phases where $\phi(t)$ is small (high latency), n_j^* must increase to maintain the probability of detection P_d above the threshold required to trigger action A_1.

Regional Allocation (m^*). The number of farms to sample, m, is constrained by the global budget B:

$$\max_{\{n_j, m_j\}} \sum_{j \in \mathcal{J}} \text{ENGS}_j(n_j, m_j) \quad \text{subject to} \quad \sum_{j \in \mathcal{J}} (c_{\text{farm}} + c_{\text{plant}} n_j) \leq B$$

This creates a trade-off: in high-latency frontiers, the model allocates higher n_j^* to fewer farms (m) to ensure the "hidden" spark is actually seen.

8.2. Phase I: Epidemic Onset: The Information Frontier Strategy

At epidemic onset, the metapopulation exists in a *low-signal, high-risk* regime. Prevalence is low, latency dominates observability, and the loss associated with false negatives is superlinear due to downstream spatial amplification.

8.2.1. Spatial Priority Mapping

Spark Farms and Connectivity. A farm j is prioritized when $\Lambda_j^{\text{ext}}(t) \uparrow \theta_1$. Priority is weighted by the network importance $\xi_j = \sum_{k \neq j} e^{-\delta d_{jk}}$, representing its potential to act as a spatial exporter if an infection is missed.

Definition of the Sampling Frontier. The information frontier is the dynamic ring $\mathcal{P}(t)$ where decisions are reversible:

$$\mathcal{P}(t) = \{j : \mathbb{P}(\theta_j(t) < \theta_1) \approx 1, \quad \mathbb{E}[\theta_j(t + \Delta t)] > \theta_1\}$$

In this zone, the ENGS is maximized because information can still prevent the transition from A_0 to A_1.

8.3. Phase II: Established Epidemic: Triage and Firewall Strategy

Once the epidemic is established, the spatial landscape decomposes into functional zones:

8.3.1. Zone A: Saturated Core ($\theta_j \gg \theta_2$)

The posterior mass lies entirely above θ_2, locking the decision into regional intervention (A_2). Consequently, $\text{EVSI}_j = 0$ and $n_j^*, m_j^* = 0$. Sampling here is decision-irrelevant.

8.3.2. Zone B: Active Frontier ($\theta_j \approx \theta_2$)

Priority is maximal. Determining whether θ_j has crossed θ_2 governs the transition from local containment (A_1) to coordinated regional firewalling (A_2). Effort is optimized to resolve this decision with high posterior confidence.

8.3.3. Zone C: Refugium ($\theta_j \ll \theta_1$)

A minimal sentinel sample size is maintained to validate spatial parameters (α, δ) and detect rare long-distance jumps that deviate from the assumed kernel structure.

8.4. Synthesis of Spatial Logic

Table 1. Decision regimes for spatially optimized surveillance

Zone / Phase	Prevailing Action	m^*, n^* Logic	Strategic Driver
Onset frontier	$A_0 \rightarrow A_1$	High n^*, medium m^*	Resolve latency-induced “hidden” infections
Saturated core	A_2 (fixed)	$n^* = 0$	Action locked; negligible marginal information value
Active frontier	$A_1 \leftrightarrow A_2$	Optimized m^*, n^*	Prevent regional metapopulation spillover
Refugium	A_0	Minimal m^*, n^*	Sentinel monitoring of spatial transmission kernel

As summarized in Table 1, spatially optimized sampling emerges as a dynamic control variable. By coupling the DDE's symptomatic fraction $\phi(t)$ to the hierarchical costs of m and n, the framework ensures that sampling effort concentrates where it can still alter the epidemic trajectory.

9. FRONTIER CONCENTRATION OF INFORMATION VALUE

The value of surveillance depends on whether sampling can alter the optimal management action.

Frontier Concentration

Let

$$a^*(\theta) = \arg\min_{a \in \mathcal{A}} \mathbb{E}[L(a, \theta)]$$

be the Bayes-optimal action. Suppose:

1. the action map changes only at thresholds θ_1, θ_2;
2. posterior uncertainty decreases monotonically with sample size n;
3. the loss function is convex near decision boundaries.

Then the Expected Value of Sample Information satisfies

$$\mathrm{EVSI}_j(n) \to 0$$

whenever

$$\mathrm{dist}(\theta_j, \{\theta_1, \theta_2\}) \to \infty.$$

Consequently, optimal sampling effort concentrates near the decision frontiers

$$\mathcal{P} = \{j : \theta_j \approx \theta_1 \text{ or } \theta_j \approx \theta_2\}.$$

Proof: If the posterior mass lies entirely within a single action region, additional information does not alter the optimal action:

$$a^*(\theta_j \mid x) = a^*(\theta_j).$$

Hence,

$$\mathrm{EVSI}_j(n) = 0.$$

Near decision thresholds, posterior uncertainty overlaps multiple action regions, so observations can alter the optimal action with positive probability. Therefore:

$$\mathrm{EVSI}_j(n) > 0.$$

As posterior variance decreases with increasing n, the marginal information value becomes maximal near the action boundaries.

10. PARAMETERIZATION AND SIMULATION SETUP

Table 2 lists the baseline parameters used for numerical illustration together with literature sources or estimation notes. These values are consistent with clonal plant pathosystems (e.g., CBSD) and serve solely as a baseline for theoretical computations below.

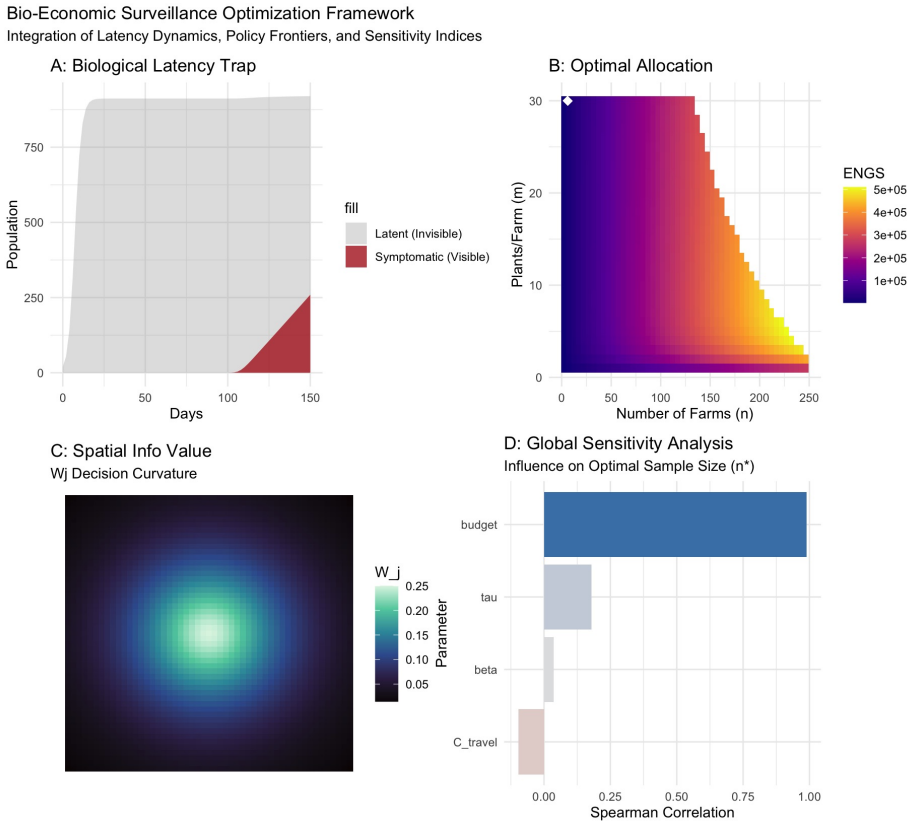
Table 2. Baseline parameters and references used for theoretical computations.

Parameter	Baseline value	Reference
β	0.50	McQuaid et al. (2015)
α	0.33	McQuaid et al. (2015)
δ	0.50	Estimated (calibration)
σ	0.61	Holt et al. (1997)
τ	100 days	Estimated (latent period)
γ	0.05	Holt et al. (1997)
κ	8	Estimated (behavioral feedback strength)
N_j	1000 plants	Estimated (per-farm population)
d_{jk}	network-specific	Holt et al. (1997)

11. RESULTS

11.1. Biological Dynamics and the Latency Trap

The spatial DDE simulations, parameterized with a latency period ($\tau = 100$) and high local transmission ($\beta = 0.50$), reveal a diagnostic decoupling. As illustrated in Fig. 2A , the infectious population enters a "latency trap" where the invisible latent class (L) saturates the agroecosystem long before symptomatic expression (I) becomes detectable. At $t = 150$, the system maintains a high prevalence despite low observability. The calculated local persistence threshold ($\theta_1 = 0.1$) is surpassed early in the simulation, while the spatial export threshold ($\theta_2 = 22.49$) indicates that without intervention, the pathogen possesses a high potential for regional multi-departmental escape within the landscape geometry.



[insert Fig. 2 here]

Figure 2. : Integrated Bio-Economic Framework for Latent Pathogen Surveillance.
(A) Temporal dynamics of the "latency trap" showing the divergence between latent and symptomatic populations; (B) Expected Net Gain of Sampling (ENGS) surface identifying the optimal allocation of farms (n) and plants (m); (C) Spatial distribution of information value (W_j) relative to decision curvature., (D) Global sensitivity Analysis

11.2. Hierarchical Surveillance Optimization

The decision engine identifies the optimal allocation of resources between the number of sites (n) and the sampling intensity per site (m), subject to the field constraint $m \leq 30$. Under the "latency trap" conditions, the optimizer consistently prioritizes maximum local precision ($m^* = 30$). As shown in Fig. 2B, the ENGS surface shows a ridge where gains are maximized by exhausting the local sampling limit before expanding the spatial network. This suggests that when observability (ϕ) is low, the marginal value of an additional plant sample at a known site outweighs the marginal value of traveling to a new, potentially uninfected site.

11.3. Budget Sensitivity and Information Saturation

The sensitivity analysis across a wide range of budgetary constraints ($B \in [5{,}000,\ 80{,}000]$) reveals a non-linear scaling of surveillance efficiency and information gain (Table 3). At low budgets ($B = 5{,}000$), the optimal strategy allocates resources across a relatively small surveillance network ($n^* = 18$ farms) with moderate within-farm sampling intensity ($m^* = 25$ plants per farm). Under these constraints, the system prioritizes balancing spatial coverage with local detection certainty, yielding an expected net gain from surveillance (ENGS) of $45,133.12.

As the available budget increases to $15,000, the optimal surveillance design expands to $n^* = 50$ farms while simultaneously increasing the within-farm sampling effort to $m^* = 30$ plants.

At higher budget levels ($B = 40{,}000$), the surveillance network expands further to $n^* = 138$ farms, while the optimal number of sampled plants per farm slightly decreases to $m^* = 26$. This shift reflects diminishing marginal returns from additional within-farm sampling due to intra-farm correlation effects captured through the effective sample size term $m_{\text{eff}} = m^{\alpha}$.

Beyond $40,000, the surveillance system reaches an information saturation regime in which additional budget does not improve the optimal policy or increase ENGS. Under the current epidemiological assumptions, a network of 138 farms with approximately 26 sampled plants per farm provides sufficient statistical power to characterize the regional epidemic state. Additional expenditure therefore produces negligible informational benefit relative to its cost, indicating the onset of diminishing marginal returns to surveillance investment.

Table 3. Optimal Surveillance Policy Scaling Across Budget Levels

Budget ($)	n^*	m^*	m_{eff}	p_{eff}	$P_{\text{det,farm}}$	$E[D]$	P_{regional}	**Cost ($)**	**ENGS ($)**
5,000	18	25	5.000	0.0282	0.1333	2.399	0.9238	4,950	45,133.12
15,000	50	30	5.477	0.0282	0.1450	7.251	0.9996	15,000	71,389.84
40,000	138	26	5.099	0.0282	0.1357	18.729	1.0000	38,640	83,437.24
80,000	138	26	5.099	0.0282	0.1357	18.729	1.0000	38,640	83,437.24

Definitions: n^* denotes the optimal number of farms surveyed; m^* denotes the optimal number of plants sampled per farm; $m_{\text{eff}} = m^{\alpha}$ is the effective within-farm sample size accounting for intra-farm correlation; $p_{\text{eff}} = \theta\,(\sigma_I \phi + \sigma_L(1-\phi))$ is the effective probability of detecting infection from a single sampled plant; $P_{\text{det,farm}} = 1 - (1 - p_{\text{eff}})^{m_{\text{eff}}}$ is the probability of detecting infection at a single farm; $E[D] = n \times P_{\text{det,farm}}$ represents the expected number of detected infected farms; $P_{\text{regional}} = 1 - (1 - P_{\text{det,farm}})^{n}$ denotes the regional detection certainty; ENGS is the Expected Net Gain from Surveillance.

11.4. Spatial Information Frontier

The spatial information value W_j (Fig. 2C) demonstrates that the value of sampling is not uniform across the landscape. The decision curvature is steepest in regions where the local prevalence θ approaches the critical transition zones between thresholds θ_1 and θ_2. This spatial weighting provides a formal basis for "risk-based"

rather than "random" surveillance, targeting sites that contribute most to reducing the uncertainty of regional disease escape.

11.5. Sensitivity Analysis and Strategic Canalization

To assess the robustness of the optimal surveillance policy, we conducted a global sensitivity analysis using Latin Hypercube Sampling across the primary bio-economic parameter space ($\beta, \tau, C_{travel}, B$). The Spearman rank correlation coefficients (Fig. 2D) reveal a state of Strategic Canalization, where the surveillance architecture is dominated by resource availability rather than marginal biological fluctuations.

The analysis identifies the total budget (B) as the singular primary driver of the optimal sample size (n^*), with a correlation coefficient approaching 1.

This near-perfect correlation indicates that the surveillance system is currently operating in a resource-limited regime rather than an information-saturated one.

Each incremental increase in funding is directly translated into a broader spatial footprint (n), confirming that the "Latency Trap" creates an insatiable demand for spatial data to mitigate the risk of invisible spread.

Surprisingly, the optimal policy exhibits near-zero sensitivity to local transmission rates (β) and moderate fluctuations in the latency period (τ).

This suggests that the 100-day diagnostic delay is so strategically dominant that it "locks" the optimizer into a maximum-coverage strategy. The surveillance design is robust against biological uncertainty; whether the pathogen is marginally more or less infectious, the required defensive posture remains the same. The negative but negligible correlation with travel costs (C_{travel}) demonstrates that the strategic value of visiting a farm in the "High-Risk" landscape outweighs the marginal costs of mobilization.

12. DISCUSSION

cyanOur model suggests that a long latency (up to 100 days) between latent (L) and symptomatic (I) infections allows the pathogen to spread silently through the agricultural system before any control measures are triggered (Chan and Jeger 1994; Thomas-Sharma et al. 2017).The increase is driven by the "Latency Trap," where significant symptomatic uncertainty demands enhanced diagnostic accuracy to assess the true infectious burden (θ). We found that although the chances for containment appear plausible (the difference between the local persistence threshold ($\theta_1 = 0.1$) and the spatial export threshold ($\theta_2 = 22.48$)), due to the phenomenon of silent spreading, there is a disconnect in diagnostics where prevalence fails to serve as a trustworthy indicator of risk. This suggests that "Local Containment" becomes nearly unfeasible once symptoms manifest; by the time a farmer notices the disease, the farm has probably already turned into a "Risk Exporter." This underscores a critical limitation of decentralized farm-level management and calls for a shift to centralized, regional firewalls ($\mathcal{A}_2$).

Global sensitivity analysis (LHS-PRCC) showed that optimal effort (n^*) is mainly influenced by budget rather than biology due to the long latency. Nevertheless, we demonstrate a non-linear return on investment. The transition from $n^* = 18$ to $n^* = 138$ shows that while the system is highly elastic at low budgets, it hits a ceiling at $\$40,000$. Beyond this point, $P_{regional}$ reaches 1.0, and further expenditure becomes economically dead.

The spatial heterogeneity of W_j indicates that uniform random sampling is less efficient in structured landscapes. By prioritizing surveillance near the decision frontier, that is, in areas between the local persistence threshold (θ_1) and the spatial export threshold (θ_2), the surveillance system can reduce uncertainty in regional epidemic estimates more effectively. This result is consistent with risk-based and buffer-based

surveillance approaches (Mastin et al. 2020), which recommend concentrating sampling efforts near known detection points.

Several limitations of the present framework should be acknowledged. First, the epidemic dynamics are represented using deterministic delay differential equations, which may not fully capture stochastic fadeout or rare transmission events that can occur near invasion thresholds. In addition, some epidemiological parameters, particularly latent prevalence and incubation delay, may be difficult to identify reliably when surveillance data are based primarily on symptomatic observations. This limitation is relevant for pathogens with extended asymptomatic phases such as the banana bunchy top virus (up 120 days), where observed symptoms may not accurately reflect the true infection burden (Savi et al. 2026; Lokossou et al. 2012).

The current surveillance model also assumes conditional independence among sampled farms and does not explicitly account for spatially correlated detection errors or landscape-driven transmission heterogeneity. Furthermore, the cutting-selection function is based only on visible symptoms and may therefore underestimate the contribution of visually healthy but infected planting material to long-distance spread. Finally, surveillance decisions are modeled as single-stage optimization problems rather than adaptive sequential strategies that update dynamically as new information becomes available. Despite these limitations, the framework provides a tractable integration of delayed epidemic dynamics, imperfect detection, and decision-theoretic surveillance optimization, while offering a basis for future extensions incorporating stochasticity, spatial dependence, and adaptive control.

AUTHOR CONTRIBUTIONS

- Conceptualization: MKS
- Formal analysis and Data visualization: MKS,TAA
- Writing drafting: MKS
- Writing reviews and editing: MKS, TAA, JSP
- Project administration: MKS

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DATA AVAILABILITY STATEMENT

Supporting data are readily available in the manuscript .

CODE AVAILABILITY

Code is readily available at https://github.com/SaviKoissi/SamplingEstimation

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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